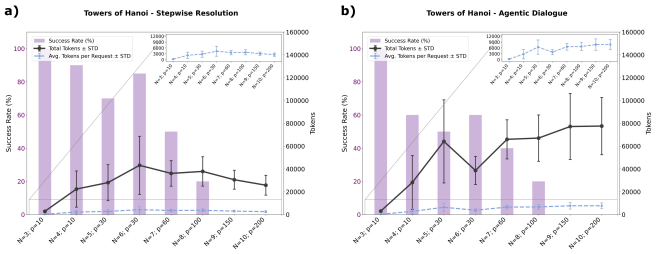

FIG. 2. Comparison of performance across two solving strategies for the Towers of Hanoi puzzle: **(a)** stepwise resolution and **(b)** agentic dialogue. Each chart displays the success rate (purple bars, left axis), the total number of tokens used with standard deviation (black line, right axis), and the average tokens per request (blue dashed line, right axis). Insets provide a zoomed-in view of the average per-request token usage. Here, N denotes the number of disks in the puzzle, and p indicates the number of moves requested per substage.

consumed to solve the complete puzzle, calculated as the cumulative token count across all substages in a trial. In contrast, the blue dashed curve reflects the average number of tokens used per substage, capturing the typical cost of a single reasoning step within the larger sequence.

The inset in the top-right corner zooms into the lower range of the blue curve, allowing better visual comparison between configurations, as per-substage token usage remains relatively small in scale. This dual analysis reveals how overall reasoning effort and localized step efficiency evolve as task complexity increases.

There is a clear rise in problem complexity as the number of disks increases, reflected in the progressive drop in success rate. When the problem reaches higher levels of difficulty (i.e., $N \geq 8$), we observe a notable decrease in the total number of tokens used by the model during reasoning. However, this trend must be interpreted with caution, as the total token count (black line) is highly sensitive to the substage at which the model makes a mistake. If the LRM fails early in the resolution—e.g., during the first substage—the overall token consumption remains low, regardless of the overall complexity of the puzzle. Conversely, mistakes made in later substages accumulate more tokens before failure occurs.

For this reason, analyzing the average number of tokens per request (blue line) provides a more stable indicator of reasoning effort. Unlike total token usage, this metric is independent of the specific substage at which failure occurs and reflects only the intrinsic difficulty of each individual substage. Since substage complexity tends to grow with the overall task, this metric offers a clearer view of how demanding each reasoning segment is, independent of earlier or later errors in the full resolution.

The evolution of token usage, together with the success rate, confirms that when puzzle complexity increases - yet remains within the LRM's solvable range - the token rate also rises accordingly, signaling increased effort to meet the challenge. However, when the puzzle exceeds the LRM's capabilities, the token rate drops, reflecting early-stage failures and an implicit recognition of task infeasibility. This supports the interpretation that the complexity of each substage directly mirrors the global difficulty of the overall puzzle.

2. *Agentic Dialogue:*

The procedure is equivalent to the previous experiment, evaluating the same configurations from $N = 3$ to $N = 10$, with 10 independent trials conducted for each (Fig. 2 *b*) presents the corresponding results.

Performance shows a clear drop compared to the stepwise resolution setup. The model begins to struggle as early as $N = 4$, clearly exhibiting worse performance in solving the problem compared to the previous methodology. However, the token usage patterns reveal a contrasting trend. Unlike in the previous experiment—where total token usage decreases sharply as complexity increases and failure occurs early—in this case, the LRM agents continue to invest considerable effort even when they are ultimately unable to solve the task. Token rates remain high across configurations.

This behavior likely reflects that the agents recognize the task as solvable within their theoretical capacity, yet fail to discover the correct sequence of actions. This interpretation aligns with empirical observations: in most trials, the agentic system maintains extended dialogues without committing illegal moves, but ends up looping indefinitely—oscillating forward and backward—without ever converging to a valid solution.

B. River Crossing

To test the capabilities of the LRM in solving this puzzle, we conduct a series of tests with varying complexity. The boat capacity is altered from $k = 2$ to $k = 4$, and the number of agent–actor couples ranges from $N = 2$ to $N = 100$, always respecting the constraints for solvable configurations. By restricting the experiments to valid instances only, we ensure that the performance reflects genuine reasoning ability, rather than limitations imposed by unsolvable setups. Each configuration is repeated 10 times (Fig. 3).

From Fig. 3, a clear trade-off between success rate and token consumption is observed. The most difficult configurations—($N = 4$, $k = 3$) and ($N = 5$, $k = 3$)—demand the highest number of tokens in the model's attempt to solve them. Conversely, the simplest configurations—those with $k = 2$ and $k = 4$ regardless of N—pose little challenge to the system, which is able to solve them without significantly increasing token usage.

The results also show that increasing the puzzle size (i.e., increasing N) does not exhibit a direct correlation with reasoning complexity: neither the number of tokens consumed nor the success rates deteriorate consis-

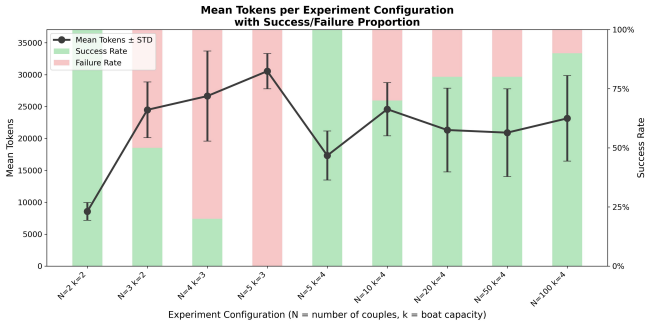

FIG. 3. Performance of the LRM in the River Crossing puzzle across different solvable configurations, defined by the number of couples N and boat capacity k. The black line with error bars represents the average number of tokens used by the model per configuration ($\pm$ standard deviation), while the colored bars indicate success (green) and failure (red) rates.

tently with larger N. Even in large-scale instances such as ($N = 100$, $k = 4$)—which require the model to perform up to 200 correct moves—the LRM solves the task reliably without significant effort or resource usage. This suggests that the challenge is not inherent to puzzle size itself.

Instead, the true difficulty appears to arise from specific configurations with a drastically reduced solution space. A particularly surprising effect emerges in configurations ($N = 4$, $k = 3$) and, especially, ($N = 5$, $k = 3$), where the system reaches a peak in resource demand while simultaneously showing the lowest efficiency.

V. DISCUSSION

The experimental results presented in this paper provide a deeper insight into the reasoning abilities of Large Reasoning Models (LRMs) when solving puzzles of increasing complexity. Furthermore, they offer a structured framework to qualitatively compare our findings with those reported by Shojaee et al. [2].

Fig. 4 illustrates this comparative analysis. It contrasts the success rate trends from our experiments with those inferred from the original study by Shojaee et al., focusing on two distinct reasoning scenarios: the Towers of Hanoi (a) and the River Crossing problem (b). It is essential to clarify that the original paper does not disclose the exact success rate values for each configuration. Therefore, the performance curves for Claude 3.7 Sonnet and DeepSeek-R1 are manually estimated from the published plots and should be interpreted as approximations.

However, this limitation is not problematic for the purpose of our study. Our goal is not to benchmark models against each other in terms of absolute performance, but rather to analyze and contrast their general behavioral trends under varying levels of task complexity and prompting strategies. This comparative visualization reveals consistent patterns across both sources, and supports broader conclusions about the cognitive capabilities and limitations of LRM-based agents when faced with structured reasoning tasks.

Related to the Towers of Hanoi experiment (see Fig. 4*a*), one of the most widely noted criticisms of the original study by Shojaee et al. [2] concerns the interpretation of failure on larger puzzles. While the authors attribute the drop in success rate exclusively to the increasing intrinsic complexity of the task, other voices in the research community suggest that these failures may also stem from technical limitations—namely [5, 6], the model's awareness of its restricted context and output windows. According to this perspective, the model may possess a conceptual understanding of the solution, yet refrain from generating complete reasoning traces due to the anticipated length of the required output.

To examine these hypotheses, we implement two decomposition strategies—stepwise resolution and agentic dialogue. In both approaches, the original problem is split into smaller, independent sub-puzzles. Instead of prompting the LRM to solve the full puzzle in a single request, each prompt focuses on a localized task: continuing from the configuration achieved in the previous sub-puzzle and making a fixed number of valid moves toward the final solution.

Fig. 4*a)* shows that, even when decomposing the task, performance does not significantly improve compared to the original study. Once puzzle complexity reaches approximately $N = 8$, both strategies fail consistently. This result suggests that the difficulty lies not in the output length or window size, but in the structural complexity of the configuration itself. Therefore, beyond limitations in output space, current LRMs still exhibit intrinsic weaknesses when attempting to solve certain classes of complex reasoning problems.

The stepwise resolution strategy also reveals a clear correlation between problem complexity and the number of tokens consumed during reasoning. As the puzzle becomes more challenging, LRMs tend to use more tokens in their reasoning process. This trend holds as long as the model perceives the task as solvable. However, when the complexity surpasses what the LRM considers tractable, token usage drops sharply. In those cases, the model refrains from investing resources in a problem it implicitly deems unsolvable. This pattern is consistent with the observations reported by Shojaee et al.

Such dynamics suggest that high token expenditure may signal the model's belief that a solution is within reach. Interestingly, the agentic dialogue approach offers an additional perspective on this behavior. Despite yielding the worst success rates in terms of accuracy, it exhibits a unique and revealing pattern. With two LRM agents interacting in alternating turns, the system does not "give up" on complex tasks. Instead, it shows the opposite trend: the more complex the puzzle, the greater the token consumption.

Unlike the stepwise or direct resolution approaches, where token usage drops when the model anticipates fail-

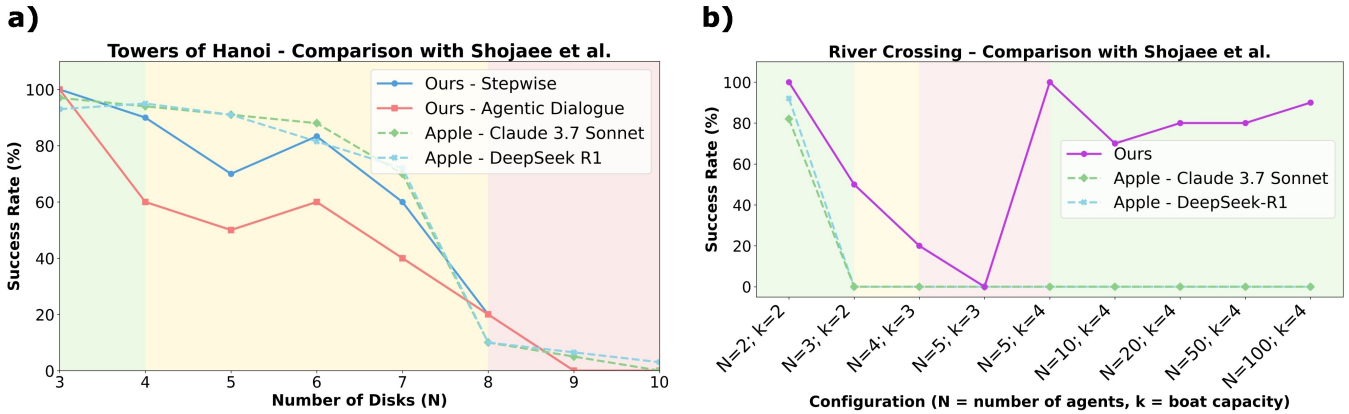

FIG. 4. Comparison of success rate trends between our experimental results and those reported by Shojaee et al. [2]. **(a)** Towers of Hanoi: stepwise prompting and agentic dialogue strategies are evaluated across increasing numbers of disks (N). **(b)** River Crossing: multiple configurations with varying number of agents (N) and boat capacities (k) are assessed. Apple's results (Claude 3.7 Sonnet and DeepSeek-R1) are estimated from the original paper, as exact success rate values are not publicly reported. This qualitative comparison highlights differences in reasoning dynamics and failure patterns, rather than direct performance benchmarking.

ure, the collaborating agents in agentic dialogue continue investing resources even when the task proves unsolvable. This persistent effort indicates that the agents internally estimate the problem as solvable, despite empirical evidence showing they are ultimately incapable of reaching the solution. They engage in extended exchanges filled with locally valid but unproductive moves—cycles that, while legal within the rules, never lead to resolution. This behavior reinforces the notion that such puzzles remain out of reach for current LRMs, not due to explicit violations of logic or game rules, but because of a deeper incapacity to structure and execute coherent long-horizon strategies.

Focusing on the River Crossing task (Fig. 4*b*), we observe a substantial shift in performance when the problem is reconfigured relative to the original study. By restricting the experiments to initial configurations with known solvable solutions, the LRM demonstrates remarkably strong reasoning capabilities. What was previously its Achilles' heel becomes one of its most reliable strengths, highlighting that an increase in problem size does not necessarily lead to reduced effectiveness. This finding reinforces the idea that LRM reasoning performance is highly dependent on the nature of the task itself and whether the model has been exposed to similar patterns during training. As a result, drawing general conclusions about their reasoning abilities proves challenging, as these systems exhibit strong context- and formulation-dependent behavior.

In the original study by Shojaee et al., all puzzles are intrinsically framed with a linearly increasing difficulty, proportional to problem size. However, unlike the Towers of Hanoi task, the River Crossing puzzle exhibits a non-linear difficulty profile. Our experiments reveal that the peak in difficulty occurs at the intermediate configuration $N = 5$; $k = 3$, rather than at the largest instances. This behavior is characteristic of combinatorial problems, where the solution space is not uniformly challenging but instead divided into distinct regions [19].

Configurations with fewer agents, such as $N < 5$; $k = 3$, have relatively few valid solutions, but those solutions are easy to discover due to their intuitive structure. On the other hand, configurations with more agents, such as $N > 5$; $k = 3$, present a broader solution space. While they require longer action sequences, they do not necessarily demand the same level of precision, making it easier for the model to eventually find a valid path. The configuration $N = 5$; $k = 3$, however, lies on the boundary between these two regimes. It contains very few viable solutions, each of which requires extremely precise reasoning and fine-grained planning. The absence of intuitive structural cues in this regime amplifies the cognitive load and makes it particularly prone to failure.

Such "phase transition" regions are notoriously difficult even for conventional search-based algorithms and are typically addressed using combinatorial optimization methods such as graph traversal, constraint satisfaction solvers, neural approximators, or evolutionary algorithms. In contrast, large language models operate through stochastic token generation, relying on learned semantic and syntactic patterns rather than systematic search mechanisms. This fundamental difference in approach makes them especially vulnerable in tightly constrained solution spaces, where even minor deviations can derail the entire reasoning process.

Looking ahead, the next steps in this project will primarily focus on scaling up the experimental framework. This includes running a larger number of tests per configuration to reduce statistical variance and expanding the set of configurations analyzed to capture a broader range of task complexities. Additionally, it would be valuable to assess whether the observed performance trends gen-